

COLIN TOFT

SOFTWARE DEVELOPER



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ColinToft



colintoft

SKILLS

Programming: C/C++, Python, Go, Java, Scala, C#, Unity, HTML/CSS/JavaScript, Scheme

Technical Skills: ROS2, Linux/Bash, MongoDB, SQL, Express, React, Next.js, LLVM, Node, Flask, GraphQL, REST APIs

EXPERIENCE

NVIDIA

Compiler Software Engineering Co-op

Austin, TX

May – August 2025

- Incoming compiler software engineering intern

University of Waterloo

Waterloo, ON

Undergraduate Research Assistantship – Social and Intelligent Robotics Research Lab

January – April 2025

- Building games in **Python**, **ROS** and **Flask** to assist in speech therapy for children, which incorporate a small humanoid robot to help with engagement

Phonic Co.

San Francisco, CA

Machine Learning Software Engineering Co-op

September – December 2024

- Built an automated model-based evaluation pipeline that reports metrics live during training (on Slack), saving significant time post-training and facilitating running ablations
- Implemented on-the-fly filtering in **Pytorch** for streaming datasets, enabling deterministic training and resumption
- Incorporated audio transcription (NVIDIA's Parakeet) and vocoder (BigVGAN) models into training and evaluation
- Downloaded and compiled a new audio dataset for training
- Built user interfaces and databases for easy human evaluations of models

Bloomberg L.P.

New York, NY

Software Engineering Co-op (Fixed Income Trading)

January – April 2024

- Developed a modular, end-to-end system in **Python/C++** allowing step-by-step replay of past client trades
- Improved memory usage in a UI panel to handle **3x** as many trades at once
- Co-developed three new automated testing suites and improved runtime to **under half** the original

Huawei Technologies

Markham, ON

Assistant Software Engineer (AI-Enabled Concurrent Program Optimization)

May – August 2023

- Innovated on a **C/C++** compiler based on the open-source **LLVM/Clang** framework
- Created an interface to query an ahead-of-time compiled AI model for compiler decision-making
- Optimized performance of a compiler pass to obtain a **62%** speedup in its execution time

University of Waterloo Blueprint

Waterloo, ON

Project Developer (Sistema Toronto)

January 2025 – Present

- Built a calendar that volunteer music teachers can subscribe to and view upcoming lessons on their own device
- Developed an admin dashboard where administrators can see teachers' lesson types and absence info

Project Developer (Feeding Canadian Kids)

May 2023 – September 2024

- Developed a meal request database using a **Python (Flask)** backend, **GraphQL** and **MongoDB**
- Built a webpage in **React** where after school programs can create meal requests to local restaurants

University of Waterloo Robotics Team

Waterloo, ON

Software Team Member

January 2022 – May 2023

- Designed software using **C++**, **ROS2** and **ros2_control** for a 6-wheeled Mars rover with a scientific instrument arm
- Developed a **CAN** interface to bridge the robot software and firmware

Radix Labs (Biotechnology Startup)

Cambridge, MA (Remote)

Software Engineering Co-op

September - December 2022

- Automated a workflow for a large pharmaceutical company which analyzes metabolism in hepatocytes
- Designed and created a parser in **Scala** that will allow scientists to create their own workflows
- Nominated spontaneously by employer for University of Waterloo **Co-op Student of the Year**

EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Computer Science (Digital Hardware & AI Specializations)

September 2021 – April 2026 (Anticipated)